

Stock Markets Analytics Zoomcamp

by [Pythoninvest.com](https://pythoninvest.com) in collaboration with [DataTalks.Club](https://datatalks.club)

Module 2

Working with Data in Pandas

Instructor: Ivan Brigida

Picture

- Business Intelligence Analyst at Google
- Educational background in Computer Science, Finance, and Data Analytics
- Founder of [PythonInvest.com](https://pythoninvest.com)
- Retail investor with a recent interest in fundamental data and options

Agenda

Answer questions from Q&A ♦ ♦ System Design Overview
Creating a Unified Dataset from Multiple Sources (Data Generation)

- Primary keys, joins, data validation

Working with Data in Pandas and Visualisation Examples

- Data types and manipulation
- Data cleaning
- Enhancing datasets with feature generation techniques: historical returns, future returns, and dummy variables for categorical or binned data

Technical Indicators

Descriptive Analysis

Visualization

By the end of Module 2, you should be able to:

- clean and type web-scraped data;
- manipulate financial time series in Pandas;
- calculate historical/future returns;
- merge stock, macro and fundamental data safely;
- build a modelling-ready dataset.

System Design View

Module 2 focuses on data processing, feature generation, and merging datasets

Module 1: Data
Sources

Module 2: Unified
Dataframe

Module 3: The
Model

Module 4: Trading
System

Module 5:
Automation

- **Output from Module 1:** A set of files or tables (if using a database), with each view closely reflecting the original data source
- **Output from Module 2:** A single “flat” file or table containing categorical and numerical features, ready for modeling

- <https://www.mutpl.com/s-p-500-pe-ratio/table/by-month>

Vote for questions or submit yours

Join at

<https://qna.dtcdev.click/r/sma>

Docs and Q&A from
Previous years [https://
datatalks.club/docs/courses/
stock-markets-analytics-
zoomcamp/](https://datatalks.club/docs/courses/stock-markets-analytics-zoomcamp/)

LINK: <https://qna.dtcdev.click/r/sma>

[2025-2026] Selected Unanswered Questions from Slido (#SMA)

Q1: Is it doable to have a full time job and getting started on the investment train ? What kind of time investment are we talking about ? In ballpark numbers ofc

- ☐ It highly depends on your background in analytics, finance, math, and programming
- ☐ Based on 2026 HA1 data (~36 submissions / 81 total), the average time commitment is 4–9 hours per 2 weeks.

2025: 2026:

Q2: Is following the current notes (2026) enough or do we need to revisit the previous materials?

- ☐ Current year notes is enough. The content is quite similar, if you have time to prepare - read slides/watch videos from 2025

Q3: Can I become a professional trader after this course?

- ☐ No. This course is more about analytics, economics, and ML, less about trading.

Q4: I want to make gold prediction. Will I be able to build a prediction application using this course ?

- ☐ Yes, you can. For example, you include daily prices for gold (and other precious metals/commodities) + technical indicators + gold-specific factors like demand/supply, US dollar strength, consumer expectations and economics stability, CB reserves, etc..

Add more questions here: <https://qna.dtcdev.click/r/sma>

<https://app.sli.do/event/59tMPiTDjYJDWqL4UaPXog/live/questions>

[1/2] Selected Comments from HA1

To update

C1: "Don't know what is required of project yet so not sure how to answer question 6". "Although open ended questions are great, but for a person with zero exposure to finance like me, I have no idea what is the way to solve Question 7"

- ☐ The course is designed in a way that you "experiment" and suggest new ideas. There are no "wrong" answers, the goal is to boost your creativity;

- Add more questions here: <https://app.sli.do/event/hrRKRfWkRkjvwfdcWb7iY6>

To update

C6: “I would omit vague or general question or formulations. I don't have a background on practical aspects of stock market and the propositions like "look at *some* data", when there are 1000 curves, indices, etc do not make a life easier. I would concentrate on some specific, most important topics and clear instructions what to read additionally. It would be great if one could formulate those free questions as the following "After reading this and this, you should be able to understand this and this". In this case the aim of reading or researching something is clear.”

- ❑ This course runs for the first time - don't expect everything to be well formulated and crystally clear, but ask questions and leave the feedback
- ❑ It's fine to leave some of the problems not answered. If it takes too much time -> feel free to skip
- ❑ The breadth of the materials and starting experience from students is huge - we can't explain everything very detailed (in fact, there are 2+ years long Masters programs on this). We'll try to provide a standard dataset/output at every stage, but we want the course to be challenging for experienced economists/traders/financists/researchers too.

- ☐ (Hard to check automatically): "I made the repository of my homework private but I have invited ... so that he can view the homeworks"
- ☐ (Links not working/ nothing there): "http://not_applicable_dot_com_please_make_it_optional"
- ☐ (Doesn't help much for 1 person reading all of that) "See Homework Url" or "Explained in the colab sheet."
- ☐ (Fair, but could you do better?) "i have no good idea by now, so I afraid have no good answer."
- ☐ (Not relevant) "Index(['Open', 'High', 'Low', 'Close', 'Volume', 'Dividends', 'Stock Splits'], dtype='object')"
- ☐ (Please don't do this) chat-gpt generated generic content

Add more questions here: <https://app.sli.do/event/hrRKRfWkIvWfdeWb7iY6>

Revising Data Sources & Introducing Data Transformations

Processing data from the web
Time series for OHLCV
New time series for macro data
EU Stats Dataset
Fundamental ratios

Initial Public Offerings (IPOs): Pandas.read_html(), stack, and process data

Code snippet 1

Read data from the web using `pd.read_html()` and build a fully typed and clean DataFrame

- Understand the columns and cast them to known data types (string, integer, datetime, float) one by one.
 - Decide how to handle errors (NaN/Null or break), and whether to drop NaNs (e.g., if “future dates of IPOs” are important).
 - Apply vectorized transformations for data processing: define new columns, if needed.
 - Perform basic exploratory analysis,
-
- <https://www.multip.com/s-p-500-pe-ratio/table/by-month>

Time series data type

Code snippet 2

Pandas Getting Started Tutorial: https://pandas.pydata.org/docs/getting_started/intro_tutorials/09_timeseries.html
Comprehensive Pandas Guide to Time Series: https://pandas.pydata.org/docs/user_guide/timeseries.html

Some useful properties of time series:

- **[Reading from String]** Valid date strings can be converted to datetime objects using `to_datetime` function or as part of read functions. `pd.to_datetime()` functionality ([examples](#)).
- **[Many date-related functions]** Datetime objects in pandas support calculations, logical operations and convenient date-related properties using the `dt` accessor (example: [timedelta](#)).
- **[Deal with missing values]** You can fill in the missing data using [interpolation](#) (`ffill`, `bfill`, `interpolate`).
- **[Error handling]** `errors="coerce"` when you don't want to deal with errors.
- **[Indexing] :**
 - A large range of dates for various offsets is pre-computed and cached under the hood, making it fast to generate subsequent date ranges (just grab a slice).

Everything is time-series in stock markets

- Fast shifting using the [shift/lag](#) method on pandas objects.
- Quick access to date fields via properties such as **year**, **month**, etc.

More Macro Indicators: Volatility and Other Asset Classes

Volatility: **VIX**; Other Asset Classes: **Gold, Oil (WTI, Brent), BTC/USD**

Code snippet 3

VIX

Gold

Brent
Oil

BTC/
USD

[2026] VIX remains relatively subdued, despite renewed geopolitical and inflation risks.💎[2026] Gold remains near record highs, supported by geopolitical uncertainty and strong safe-haven demand.💎[2026] Brent oil has surged back above \$100, driven by Middle East tensions and supply disruption risks.💎[2026] Bitcoin remains highly volatile, trading around \$78K after a significant pullback from its 2025 highs.

- <https://www.multpl.com/s-p-500-pe-ratio/table/by-month>

Macro indicators: “European Union stats”

Euro Area Yield Curves

Code snippet 4

“A yield curve is a representation of the relationship between market remuneration rates and the remaining time to maturity of debt securities. A yield curve can also be described as the term structure of interest rates. The ECB publishes several yield curves, as shown below (in the Web UI)”

2024 vs. 2025

2025 vs. 2026

- [Web representation](#) vs. Python downloaded data (2 charts for 2024 and 2025)

- <https://www.mutpl.com/s-p-500-pe-ratio/table/by-month>

Fundamental Indicators for a “Traditional” Ratio Analysis

Financial stats/ratios are used to predict Enterprise Value or the “fair” value of a stock:
EPS, Revenue, Net Income, Profit Margin, EBITDA, P/E, P/B, D/E, and more.

Code snippet 5

- The **last four years** of earnings data are available on Yahoo Finance. Note it should be joined only when published (i.e., on the “Earnings date”)
- OR, you can apply web scraping: <https://www.wsj.com/market-data/quotes/WMT/financials/annual/income-statement> (for longer period stats)
- Compare with any Web UI (e.g., on [Gurufocus.com](https://www.gurufocus.com)) - what are the important factors? Read analysts’ notes or books on this.
- **BIG 7 fundamental ratios comparison:** <https://seekingalpha.com/comparison/9fdff4dc86-Magnificent-7-Stocks>
- “Ratio analysis” for more examples of ratios: <https://www.investopedia.com/terms/f/fundamentalanalysis.asp>

- <https://stackoverflow.com/questions/55940952/how-to-get-quarterly-income-statement-with-yahoo-fin>

Creating a unified dataset

to be used as an input for next steps

Candlestick chart for OHLCV data

Useful visual representation of time trends; easy to view growing (“bullish”) vs. declining (“bearish”) days

Code snippet 6

- Candlestick is the most common financial visualization of OHLCV stats (Open-High-Low-Close-Volume).
- The Plotly library allows you to create it with a few lines of code.
- You can add more line or indicators to the chart for visualising your strategy.

Simple Trading Strategy on Tech. Indicators (RSA mean reversion)

!Example strategy, not a recommendation! : Buy when the market is oversold, sell when it becomes overbought.
Buy: RSI < 30, **Sell:** RSI > 70

- **RSI (Relative Strength Index)** is a momentum indicator that measures how strongly an asset's price has moved up or down over a given period, typically **14 days**.

Link: <https://medium.com/@slisowski/backtesting-rsi-trading-strategy-in-python-37a2ecf48dc3>

Simple Trading Strategy on Tech. Indicators (SMA 10/20)

Visualisation helps to understand and refine the strategy. One potential strategy:

Buy when green (SMA10) is lower and intersects red (SMA20), sell when green intersects red again from the upper side

- SMA 10 = “Simple Moving Average 10 days”, SMA 20 = “Simple Moving Average 20 days”

Link:

- <https://www.tradingview.com/chart/EURUSD/TXANiPQP-Simple-Trading-Strategy-10-20-SMA/>

Main OHLCV dataset for 33 stocks: ~200k rows of data (39Mb)

We'll use 11 stocks from US, 11 from EU, 11 from India - you can extend this to 100, 1000, or more stocks

Code snippet 7

- The primary key (for joins) is <Date, Ticker> or just <Date> (when you build a model for one stock).
- **TO_PREDICT**: growth_future_30d (regression) or **is_positive_growth_30d_future (binary models)**.
- **CLEAN NUMERICAL FEATURES FOR MODELS**: log(Volume), growth_*, SMA10/20, growing_moving_average, high_minus_low_relative, volatility.
- Some other features to be used are **CATEGORICAL**: you need to generate “dummies” for them.
- You can add many more indicators from technical analysis (next slide) and other sources.

Historical growth is the main “memory” of OHLCV time-series to be used as a feature for future growth prediction

- Don't use absolute numbers of 'Close' as features; these time series may be always growing ("non-stationary"). The data used for training vs. test/validation shouldn't be very different. There is a potential problem if you train on "low" values of 'Close', and try to predict for "high" values (later in time)
- Historical growth is stationary, and distributed around 1.

<https://www.visualcapitalist.com/visualized-how-long-does-it-take-to-double-your-money/>

Future returns: that's what we try to predict (for 1,5,30,... periods)

1 day

30 days

- The distribution is almost 50/50, which makes it hard to differentiate sometimes (especially when there is only small growth or decline)
- You can try to predict actual growth in percentage terms using regression, but it is more difficult and not strictly necessary for making buy/sell decisions

Technical indicators - generated fields from OHLCV data

Volume, price patterns, volatility indicators: 120+ additional fields

Code snippet 8

- [PythonInvest's] article about technical indicators for hourly stocks+crypto : <https://pythoninvest.com/long-read/tech-indicators-on-crypto-and-stocks>
- Shared code for the article: [LINK](#)
- If you want to understand the topic in depth, there are plenty of books written about it: <https://www.investopedia.com/articles/personal-finance/090916/top-5-books-learn-technical-analysis.asp>

Combining all together: merge on primary keys with validation

The goal is to have one flat table with ~200-300 features and up to several million rows.

Code snippet 9

Start from **merged_df_with_tech_ind** and add all stats daily, creating **m2** to **m16** DataFrames:

dax_daily_to_merge, snp500_to_merge, Dji_daily_to_merge, Epi_etf_daily_to_merge, Gdppot_to_merge, Cpilfesl_to_merge, fedfunds, dgs1, dgs5, dgs10, vix_to_merge, gold_to_merge, crude_oil_to_merge, brent_oil_to_merge, btc_usd_to_merge

16 steps

- Start from basic OHLCV daily data (only **37 Mb** of data, 26 fields).
- Add technical indicators (left join).
- Add macro/other input sources (left joins). **Final dataset is 316MB and 203 fields.**
- **Ensure there are no duplications (add validation + check total row count). Make sure to use the ffill() method to fill missing values (NA) where available.**
- Save to file (parquet is better for pandas) or to a database.

RECAP

Data processing for web-scraped data
Time series processing for OHLCV
More macro examples
Fundamental indicators
Technical indicators
Merging everything together

ONE DATASET WITH ALL STATS (33 stocks, 203 features, 230k records, 300Mb storage in pandas / 130 Mb compressed with Brotli)

What have we learned?

Module 2

→ **HA2:** 5 questions

→ **Deadline** to submit is 2 weeks from this Monday (28 September, 11:59pm GMT+1, Dublin time)

→ **Designed** to be straightforward and relatively easy. You can write feedback in Telegram (under the HA announcement) or when you submit the HA.

→ **Get extra points** for posting your homework and tagging PythonInvest.com and Datatalks.club on any public platform (e.g. Medium, LinkedIn, X (Twitter), etc.).

Still have questions? Write FAQ ([SMA FAQ](#) vs. [Data Engineering FAQ](#)), ask in Slack, ask Ivan in Slido